# Causal Machine Learning assignment

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## Introduction

In this project the effect of eligibility for enrolling in a 401(k) plan on net financial assets is investigated.

## 1a) Effect estimation with Causal Forests

In the first phase Causal Forests were used to estimate the effect of eligibility for enrolling in a 401(k) plan on net financial assets, in function of age, income, family size, years of education, a married indicator, a two-earner status indicator, a defined benefit pension status indicator, an IRA (individual retirement account) participation indicator, and a home ownership indicator.

The causal forest analysis reveals a causal effect of eligibility for enrollment in a 401(k) plan on accumulated net financial assets. The estimated Average Treatment Effect (ATE) equals 7799.81, indicating that individuals who are eligible for a 401(k) plan accumulate, on average, approximately \$7799.81 more in net financial assets than non-eligible individuals. The estimated standard error is \$7799.81, resulting in a 95% confidence interval of [\$5477.19, \$$1.012243 \times 10^4$]. Because the confidence interval is positive and larger than 0, we conclude that being eligible for the 401(k) has a significant and positive effect on the net financial assets of a person.

In addition to the average treatment effect, the Conditional Average Treatment Effect (CATE) was examined to investigate heterogeneity in treatment effects across individuals. Figure @ref(fig:fig-cate-hist) presents a histogram of the predicted CATE values. Most individuals have positive CATE values, although negative values are also observed. Furthermore, the estimated CATE was plotted against income (Figure @ref(fig:fig-cate-unc)), revealing a clear upward trend: the effect of 401(k) eligibility appears to increase with income. Thus, becoming eligible does have a positive impact on financial assets, but mainly so for richer individuals.

To better characterize this relationship, both a linear regression model and a smoothing spline regression (with $\lambda = 1$) were fitted. The corresponding trends are displayed in Figures @ref(fig:fig-cate-inc-smooth) and @ref(fig:fig-cate-inc-lm) While the estimated curves differ somewhat at higher income levels, this part of the income distribution contains relatively few observations, implying considerably greater uncertainty in this region.

To assess the stability of the inverse probability weights, the distribution of the estimated propensity scores was also investigated using a histogram (Figure @ref(fig:fig-A-hist)). 0% of the scores were smaller than 0.01 or larger than 0.99, and only 0.02% of the scores were smaller than 0.05 or larger than 0.95. Overall, the propensity scores do not exhibit problematic extremes. In addition, a clear relationship between income and the propensity scores was observed (Figure @ref(fig:fig-A-inc)). This finding is inconsistent with the assumption of exchangeability conditional on income alone. Together with the limited number of extreme propensity scores, this provides an additional reason not to apply propensity score truncation. Removing observations with extreme propensity scores would likely shift the estimated ATE toward the treatment effect for higher-income individuals, thereby reducing the representativeness of the overall estimate. This can be viewed as an example of the bias–variance trade-off. Although trimming extreme propensity scores can reduce the variance of the estimator by limiting the influence of highly weighted observations, it may simultaneously introduce additional bias by systematically excluding specific subpopulations from the analysis.

To estimate the propensity scores and outcomes, 5-fold cross-fitting was applied. The modeling was performed using Regression Forests, as described in the documentation of the GRF package (https://grf-labs.github.io/grf/reference/causal_forest.html).

# 1b) Possible problems when the analysis additionally conditions on other variables

It was previously stated that eligibility for a 401(k) could be taken as exchangeable conditional on income. However, conditioning on additional variables can have important implications depending on their role in the causal structure.

## Role of Additional Variables in a Causal Diagram

In addition, including variables in an analysis can have different effects depending on their position in the causal diagram.

- *Collider*
  In the case of a collider *(Eligibility → Collider ← Outcome)*, conditioning on this variable introduces collider bias.
  Conditioning on a collider opens a backdoor pathway, making the estimated effect of eligibility biased and very difficult to interpret.
- *Mediator*
  In the case of a mediator *(Eligibility → Mediator → Outcome)*, part of the effect of eligibility is captured through an indirect effect (via the mediator).
  Including a mediator is therefore not incorrect, but it implies that the estimated effect is no longer the total effect, but only the direct effect.
- *Common Cause*
  In the case of a common cause *(Eligibility ← Common Cause → Outcome)*, it is appropriate to include this variable in the analysis. If a common cause is omitted, a backdoor pathway remains open, making the estimated effect of eligibility not credible. We note that conditional exchangeability given only income does not hold if there exist common causes of Eligibility and the outcome that are not controlled for.
- *Variable Affecting Only Eligibility*
  If a variable only affects eligibility, including it in the analysis does not introduce bias.
- *Variable Affecting Only the Outcome*
  If a variable only affects the outcome and not eligibility, it can be included in the analysis.
  Including such variables may improve the efficiency of the estimation.

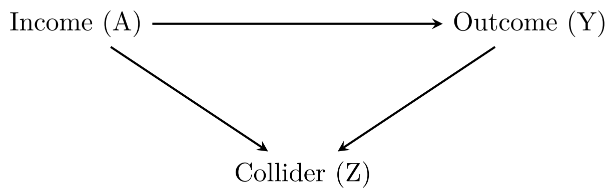
## Examples

- *Age* could potentially be a *common cause*.
  Older individuals are generally more likely to be eligible for a 401(k) program (stable full-time jobs, longer job tenure), while older people have had more time to accumulate wealth and financial assets over the course of their lives.
- *IRA* could act as a *collider*, since individuals who are ineligible for a 401(k) may substitute toward an IRA as an alternative retirement savings vehicle, while wealthier individuals with higher net financial assets are also more likely to participate in an IRA.

- *Years of Education* could be a *common cause*.
  Individuals with higher levels of education tend to accumulate greater net financial assets and are also more likely to be eligible for a 401(k) plan.

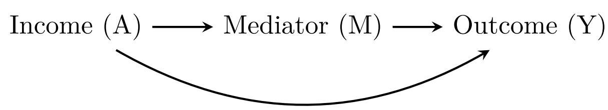
% — COLLIDER —

**Collider**



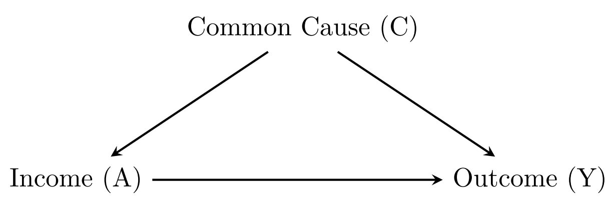

% — MEDIATOR —

**Mediator**




% — COMMON CAUSE —

**Common Cause**




## 2) Investigating heterogeneity based on subgroup analysis

Based on the predictions obtained from the causal forest, the data were split into four subgroups to investigate treatment effect heterogeneity.

First, we can see that the estimated treatment effect is positive in all four quartiles. This implies that, on average, 401(k) eligibility has a positive impact on net financial assets across the full distribution.

In terms of treatment heterogeneity, the estimated effects for the bottom three quartiles are relatively similar, ranging from \$3729.07 to \$8009.43. However, for the top quartile of individuals, which were predicted to benefit the most, the estimated effect increases to approximately \$$1.322684 \times 10^4$, with a 95% confidence interval of [\$6706.86, \$$1.974682 \times 10^4$].

This suggests that the treatment effect is not homogeneous across households, as the confidence intervals do not overlap for all subgroups. However, because the confidence intervals across quartiles still overlap, this should be interpreted as suggestive evidence of treatment-effect heterogeneity rather than definitive proof. The subgroup with the highest predicted CATE likely consists of households with characteristics such as higher income or stronger saving capacity, for whom the tax-advantaged saving mechanism is more effective.

First, we find that the estimated treatment effect is positive in all four quartiles. This implies that, on average, 401(k) eligibility has a positive impact on net financial assets for the complete distribution. However, this treatment effect differs across subgroups. The estimated ATE increases across the four predicted-CATE quartiles: approximately \$3729 in Q1, \$6236 in Q2, \$8009 in Q3, and \$$1.3227 \times 10^4$ in Q4. This already suggests that there is meaningful heterogeneity in treatment effects.

At the same time, the confidence intervals of adjacent quartiles overlap, so there is no evidence for a difference between these quartiles. However, the confidence intervals for the lowest and highest predicted-CATE quartiles do not overlap, which indicates a clear difference between individuals predicted to benefit least and those predicted to benefit most, "that even the difference between the most extreme quartiles should be interpreted cautiously")'. Overall, this thus impliestreatment-effect heterogeneity: 401(k) eligibility appears to increase net financial assets for all groups, but the estimated gains are substantially larger for the subgroup with the highest predicted treatment effects.

This subgroup most likely consists of households with higher saving capacity (e.g. higher income/ other financial characteristics,...). Therefore, this tax-advantaged saving mechanism appears to be especially effective for households that are already predicted to benefit more from 401(k) eligibility.

# 3) Investigating heterogeneity using the Efficient Influence Function

The Efficient Influence Function (EIF) was used to construct a debiased estimator for assessing treatment effect heterogeneity.

As CATE variance estimate (and its confidence intervals) are very large; ($\hat{\sigma}^2 = 8.1978899 \times 10^6$, 95% CI $[-6.8996527 \times 10^7, 8.5392307 \times 10^7]$), we also inspect what drives this value. For this purpose, we inspect whether this result is driven by a few observations with extreme EIF values, which we define as values in the top percent of efficient influence function values. The largest absolute influence-function value equals $2.1858792 \times 10^{11}$, while the 99th percentile equals $8.1554129 \times 10^9$.

These extreme EIF values can originate from three 'sources'. First, they may come from large deviations of the estimated CATE from the ATE. Second, they can come from extreme propensity scores. Third, they can be the result from large outcome residuals. Hence, we compare these for the top 1% vs top 99% of the population.

As shown in Table X, the extreme EIF values are the result of large residuals. The mean absolute residual among the top 1% most influential observations is $2.7122637 \times 10^5$, compared with $1.579485 \times 10^4$ for the rest of the sample. These extreme values for these observations are the result of higher net financial assets on average: $2.5911691 \times 10^5$, compared with $1.559544 \times 10^4$. This indicates that several high-asset households have observed outcomes that are substantially larger than predicted by the outcome model. On the other hand, the CATE-ATE deviations are also large, but smaller. The propensity scores among these extreme EIF values are rather normal.

Overall, this suggests that the wide confidence interval is not primarily caused by severe overlap violations, but by a small number of high-asset observations with very large outcome residuals. Since the debiased estimator is not constrained to be non-negative in finite samples, these large correction terms can dominate the positive squared CATE-deviation term. Therefore, the evidence for treatment-effect heterogeneity should be interpreted cautiously.

Extreme values for these influence functions can either be driven by the ... term, the propensity scores or the residuals. Therefore, we also inspect whether our extreme influence function values are driven by these. As we can see in table x, these are mainly driven by very large residuals. This is because certain observations have very large residuals, as their net financial assets is substantially larger than what's predicted. For example,...

(The 95% confidence interval still overlaps with 0, so there is no evidence of effect heterogeneity.)

To assess treatment effect heterogeneity, the variance of the CATE was calculated as given in the assignment. This gives us an idea of the extent as to which the impact of 401(k) eligibility on the net amount of financial assets is different across households with different covariate characteristics. If the effect is close to 0, then we would conclude that the treatment effect is homogenous for households characterized by similar covariates.

More formally, we estimated:

$$\sigma^2 = Var\{E(Y^1 - Y^0 \mid L)\},$$

where $Y^1$ and $Y^0$ give the outcomes under treatment (eligibility) and control (non-eligibility), respectively, and $L$ gives the covariates:

$$\tau(L) = E(Y^1 - Y^0 \mid L),$$

Thus, we can write:

$$\sigma^2 = Var\{\tau(L)\}.$$

If

$$\psi = E\{\tau(L)\}$$

gives the average treatment effect. Then,

$$\sigma^2 = E\left[\{\tau(L) - \psi\}^2\right].$$

Thus, $\sigma^2$ measures the variance of the CATE around the ATE.

Normally, we would just compute the variance around the CATE's. However, CATEs are estimated and therefore, this could lead to incorrect inference. Therefore, the efficient influence function for $\sigma^2$ is used, denoted by:

$$\begin{aligned}\phi(O) =& \{\tau(L) - \psi\}^2 - \sigma^2 \\ &+ 2\{\tau(L) - \psi\}\left\{\frac{A}{P(A=1 \mid L)} - \frac{1-A}{P(A=0 \mid L)}\right\}\{Y - E(Y \mid A, L)\}.\end{aligned}$$

The efficient influence function has mean zero at the true value of the parameter, meaning that

$$E\{\phi(O)\} = 0.$$

This property can be used to construct an estimator for $\sigma^2$. In practice, we replace the unknown nuisance functions by estimates and solve the empirical analogue of this mean-zero condition:

$$\frac{1}{n}\sum_{i=1}^{n} \hat{\phi}_i = 0.$$

Starting from the efficient influence function, this means solving

$$\frac{1}{n}\sum_{i=1}^{n}\left[\{\hat{\tau}(L_i) - \hat{\psi}\}^2 - \hat{\sigma}^2 + 2\{\hat{\tau}(L_i) - \hat{\psi}\}\left\{\frac{A_i}{\hat{p}(L_i)} - \frac{1-A_i}{1-\hat{p}(L_i)}\right\}\{Y_i - \hat{m}(A_i, L_i)\}\right] = 0.$$

We replace the (unknown) nuisance functions by empirical estimates

$$\hat{\tau}(L_i)$$

for the estimated CATE from the causal forest,

$$\hat{\psi}$$

for the estimated average treatment effect,

$$\hat{p}(L_i) = \hat{P}(A_i = 1 \mid L_i)$$

for the estimated propensity score,

$$\hat{m}(A_i, L_i) = \hat{E}(Y_i \mid A_i, L_i)$$

for the estimated outcome regression.

This gives the following debiased estimator:

$$\hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^{n} \left[ \{\hat{\tau}(L_i) - \hat{\psi}\}^2 + 2\{\hat{\tau}(L_i) - \hat{\psi}\} \left\{ \frac{A_i}{\hat{p}(L_i)} - \frac{1 - A_i}{1 - \hat{p}(L_i)} \right\} \{Y_i - \hat{m}(A_i, L_i)\} \right].$$

The first term,

$$\{\hat{\tau}(L_i) - \hat{\psi}\}^2,$$

gives the estimated variation in the CATE around the ATE. The second term is a debiasing correction based on the residuals from the outcome regression and inverse probability weighting. This correction accounts for potential misspecification in the nuisance functions.

Then, we estimated the influence-function values as

$$\begin{aligned} \hat{\phi}_i =& \{\hat{\tau}(L_i) - \hat{\psi}\}^2 - \hat{\sigma}^2 \\ &+ 2\{\hat{\tau}(L_i) - \hat{\psi}\} \left\{ \frac{A_i}{\hat{p}(L_i)} - \frac{1 - A_i}{1 - \hat{p}(L_i)} \right\} \{Y_i - \hat{m}(A_i, L_i)\}. \end{aligned}$$

The standard error is estimated as:

$$\widehat{SE}(\hat{\sigma}^2) = \frac{sd(\hat{\phi}_i)}{\sqrt{n}},$$

and a 95% confidence interval was constructed as

$$\hat{\sigma}^2 \pm 1.96 \times \widehat{SE}(\hat{\sigma}^2).$$

Ultimately, we inspected the estimated influence-function values and the inverse probability weights as diagnostics, to see if our estimates and CI's were driven by those.

#Main conclusion

Our estimate for the variance is approximately −6.68×10ˆ6 (se=…), with a confidence interval around it of … and …. As ˆ2 denotes the variance, this obviously cannot be negative in reality. However, as we use the debiased estimator, this is not constrained to be negative. To see this, consider that the first term of the EIF should always be positive (as it is squared), however if the correction term is larger than the first term, then the found estimate can be negative.

Our estimate of … is thus positive.

The 95% confidence interval includes 0, and this thus means that we cannot concldue that there is significant heterogeneity in the impact of being eligible for 401(k) schemes for different households.

As our estimates are rather large, we also took a deeper look at the values for the efficient influence functions. This check indicated that certain observations had large absolute influence function values. More specifically, these values are mostly driven by households with a high amount of net financial assets (and thus large residuals). We further investigated whether the propensity scores of these observations were close to 0 or 1, and this was not the case. Thus, these values were caused by the very large residual values caused by the right-skewed distribution, not by the propensity scores.

Overall, as the confidence intervals overlap with 0, we cannot conclude treatment effect heterogeneity. However, as these confidence intervals remain very broad, it is also difficult to conclude that there is no heterogeneity. Therefore, it is difficult to take conclusions, as inference is impacted by specific high-asset households.

# 4) Estimation of the causal effect using a orthogonal R-learner

The causal forests that were used in the previous phases have the drawback that they learn the causal effect conditional on many variables, making it both difficult to learn with good accuracy and difficult to visualize. In this fourth phase, we therefore estimate the average effect of eligibility for enrollment in a 401(k) plan on net financial assets conditional only on income. For this purpose, an orthogonal R-learner is used, controlling for the same set of covariates as before to ensure valid confounding adjustment. While both DR learner and the R-learner both are theoretically sound to use in this case, it was ultimately decided to implement the R-learner. The R-Learner was chosen because it is more robust to the local overlap failures visible in the propensity score plot. Where $\hat{p}$ is extreme, the DR-Learner amplifies errors through IPW division and requires explicit clipping, introducing bias. The R-Learner instead naturally downweights those observations via W_R = $(A - \hat{A})^2$, requiring no clipping at all.

We estimated the CATE of eligibility for a 401(k) plan on net financial assets using an R-Learner. The nuisance components, such as the conditional outcome model and the propensity score, were estimated using random forests with all covariates to improve confounding adjustment. In the final stage, the R-Learner pseudo-outcomes were modelled on income only using an ordinary weighted linear model. Compared to the approach used in question 3, where the model gives us the existence of the heterogeneity and attempts to quantify this value, the R-Learner tells how this heterogeneity varies with income, giving a more informative and directly answers the research question about the income-dependent treatment effect. The chosen R-Learner introduces an additional modelling assumption in the final stage, compared to the causal forest approaches. Specifically, by regressing the pseudo-outcomes linearly on income, the method assumes that the treatment effect varies approximately linearly with income. In contrast, causal forests estimate treatment heterogeneity fully nonparametrically, allowing for more flexible nonlinear patterns and interactions. The advantage of using a linear model in the final stage is improved interpretability and easier visualisation of the treatment effect as a function of income. If instead of linear models, other models would have been used in the approach, then the interpretability of the relationship between income and the treatment effect would have been more difficult, and would add other modeling assumptions. However, using a linear model also introduces the risk of model misspecification if the true relationship is nonlinear, which could lead to biased estimates of the treatment effect heterogeneity.

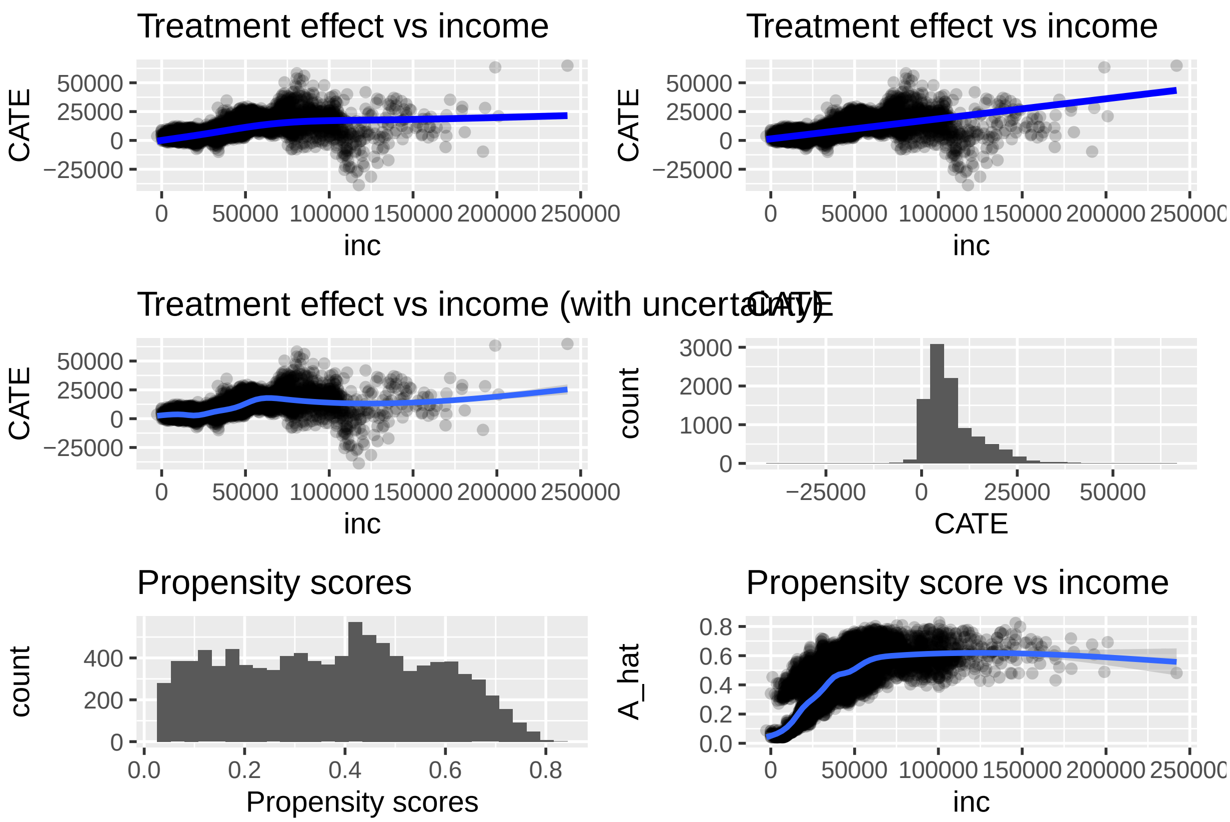
When comparing the resulting CATEs between the R-learner and the previously used causal forest approach, we could observe a similar trend, with mmost individuals having positive CATE values, and some negative values are also observed. However, the CATEs resulting from the R-learner approach resulted in a more condensed histogram with most CATEs resulting in the same peak. However, similar to the causal forest result, we can still observe a tail reaching higher CATE values. Averaging over the CATEs from the R-learner gave us an average treatment effect of 8130.6298003 with a standard error of 51.5309338. This showed that a highly stable estimate of the Average Treatment Effect (ATE) could be derived from the R-learner's individualized predictions. This could also be seen when comparing the CATEs from the different folds from the cross-fitting in terms of the income. Here, we see that the trend in all 5 folds crosses a single point equal to the average. Next to this, all five folds follow a similar linear trend. On this plot it could be observed that a high-income points with large CATEs, can realy influence the linear trend build into the R learner. This pattern implies that the R-learner's residualization step effectively removes confounding, allowing the treatment effect to emerge cleanly as a function of income. The intersection near the mean ATE is a visual confirmation of model calibration. The distribution of the predicted outcomes (E[Y]) shows a pronounced right-skew, with most observations concentrated near zero and a long tail extending toward higher expected values. This indicates that the majority of individuals are predicted to have relatively low expected responses, while a smaller subset exhibits substantially higher predicted outcomes. Such a pattern suggests that the model successfully captures heterogeneity in outcome potential, though the skewness may point to sensitivity to outliers or nonlinear effects that could be explored further through transformation or trimming. The estimated propensity scores display a fairly uniform distribution between 0 and 0.8, with no substantial clustering near 0 or 1. This confirms that the overlap (positivity) assumption holds: both treated and control units share similar covariate support, and no individuals have near-deterministic treatment assignment. Consequently, inverse probability weighting and related estimators are expected to remain numerically stable.

Together, these diagnostics demonstrate that the R-learner framework operates under favorable conditions. The outcome model captures meaningful variation across individuals, while the treatment assignment model maintains adequate overlap. This balance supports the reliability of the estimated Conditional Average Treatment Effects (CATEs) and the derived Average Treatment Effect (ATE), ensuring that causal estimates are both interpretable and robust

# Addendum:

## Q1:

```
## `geom_smooth()` using formula = 'y ~ x'
## `geom_smooth()` using method = 'gam' and formula = 'y ~ s(x, bs = "cs")'
## `geom_smooth()` using method = 'gam' and formula = 'y ~ s(x, bs = "cs")'
```




### Q2:

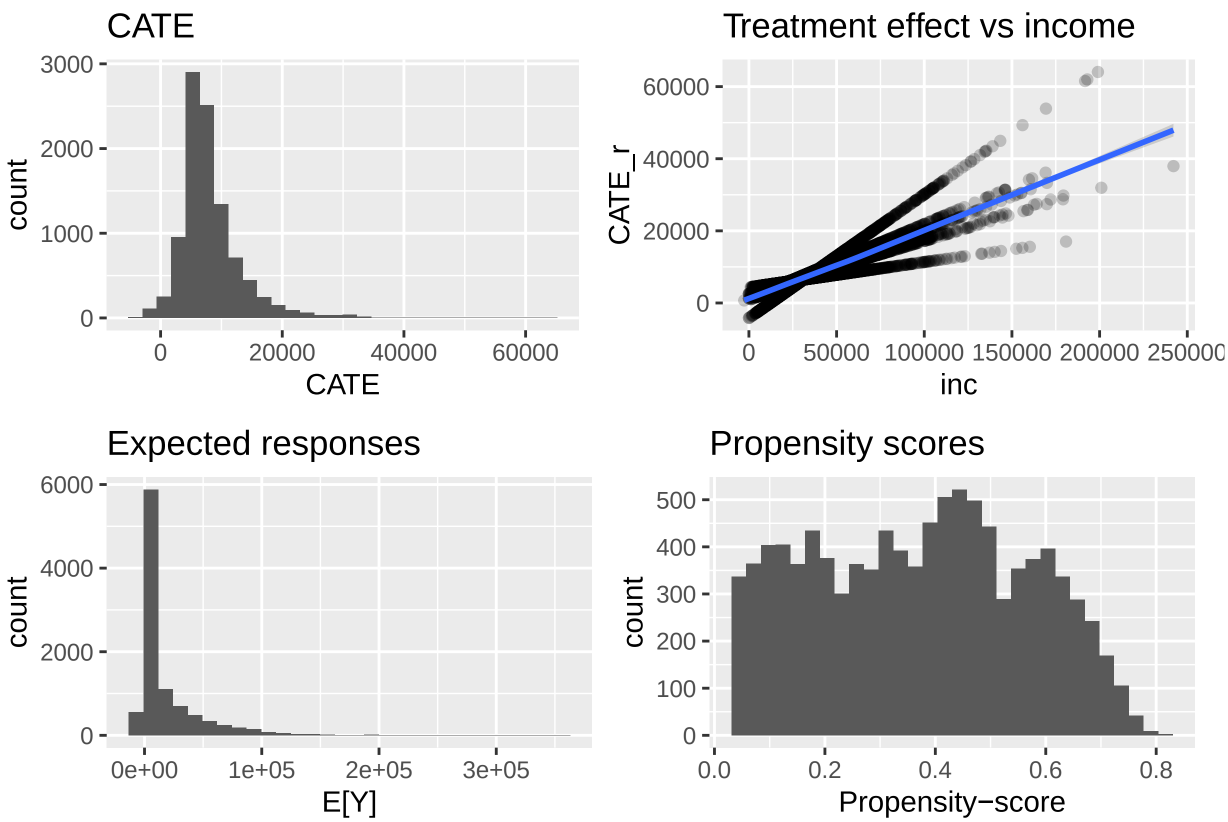
table:

Table 1: ATE estimates by predicted CATE quartile

| Quartile | Estimate | SE | CI_low | CI_high |
|---|---|---|---|---|
| Q1 (lowest predicted CATE) | 3729.07 | 1165.49 | 1444.72 | 6013.43 |
| Q2 | 6236.09 | 2371.53 | 1587.90 | 10884.28 |
| Q3 | 8009.43 | 2100.33 | 3892.79 | 12126.07 |
| Q4 (highest predicted CATE) | 13226.84 | 3326.52 | 6706.86 | 19746.82 |

### Q4 figures:

```
## `geom_smooth()` using method = 'gam' and formula = 'y ~ s(x, bs = "cs")'
```


CATE
count
CATE
Treatment effect vs income
CATE_r
inc
Expected responses
count
E[Y]
Propensity scores
count
Propensity−score